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EDA Summary

1. **Dataset Summary**
   Our dataset contains acoustic features extracted from audio recordings to better classify vocal techniques.
   a. Target Variable: Vocal Technique (belt vs. breathy)
   b. Total Observations: 500 entries → reduced to 475 after cleaning
   c. Features: Four continuous numerical features
      i. Fundamental Frequency f_0
      ii. Spectral Centroid
      iii. Spectral Rolloff
      iv. Harmonics-to-Noise Ratio (HNR)

2. **Data Preprocessing** (performed using pandas)
   a. Missing values:
      i. f_0 ⇒ 12 values
      ii. Spectral Centroid ⇒ 5 values
   b. 8 duplicate rows were found and dropped to prevent data leakage and bias

3. **Descriptive Statistics and Correlations**
   Table 1: Feature Statistics by Vocal Technique

=== OVERALL DESCRIPTIVE STATS ===

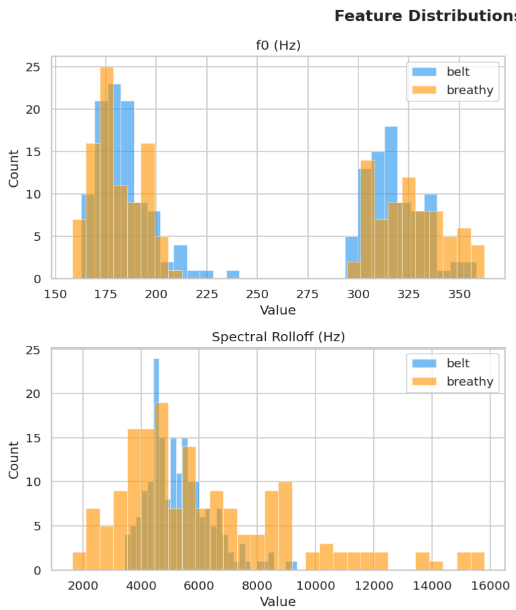
	f0	spectral_centroid	spectral_rolloff	hnr
count	349.000	349.000	349.000	349.000
mean	245.284	3095.083	5741.023	-2.762
std	71.042	749.906	2235.837	8.357
min	158.327	1420.763	1634.101	-19.713
25%	177.833	2552.408	4415.394	-8.814
50%	198.220	2985.943	5204.996	-1.904
75%	316.780	3490.785	6510.524	2.623
max	362.320	6092.156	15793.111	23.766

The statistics shown in Table 1 reveal significant differences in the acoustic profiles of the two techniques. For example, “Belt” vocals exhibit a higher mean Spectral Centroid (brightness) and clarity over time (HNR) compared to the “Breathy” vocals.

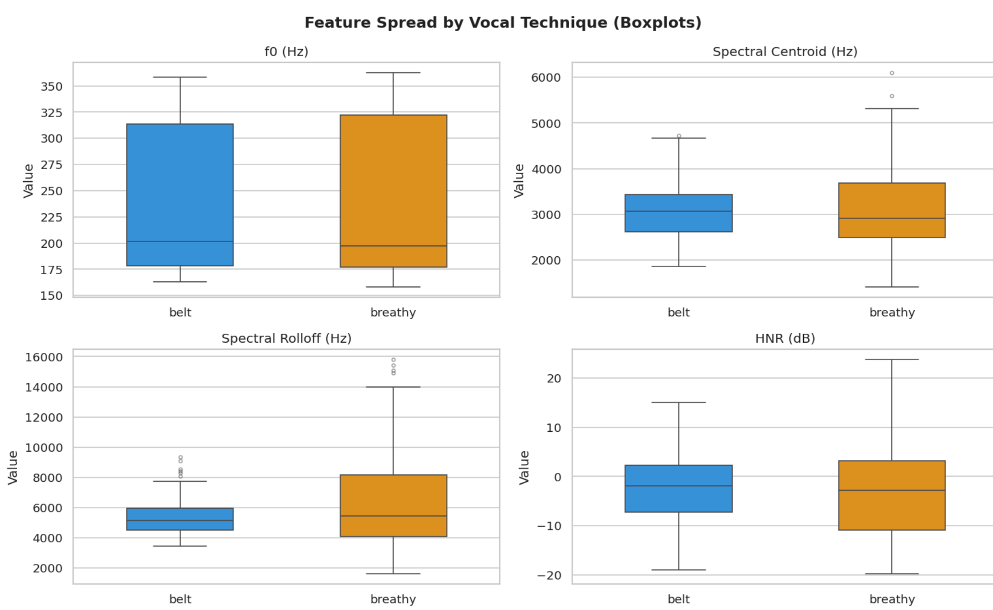
The Pearson Correlation Matrix used to calculate these statistics confirms that there is a very strong positive correlation of 0.89 between the Spectral Centroid and Spectral Rolloff features. It also shows that moderate correlations exist between f_0 and spectral features to about 0.46, while the HNR shows a moderately negative correlation with Spectral Rolloff at -0.33.

4. Visualizations

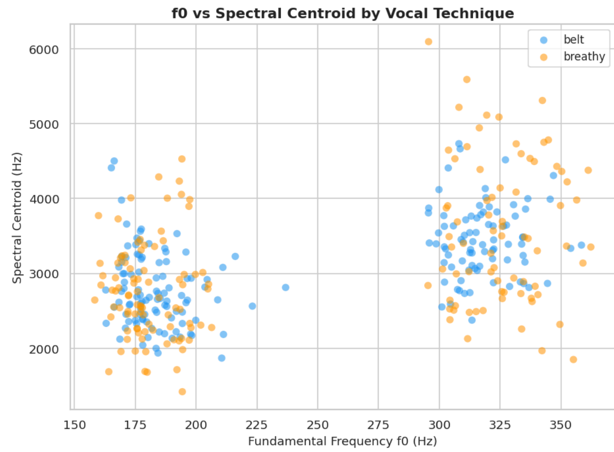
a. Histograms to show that while the HNR for “belt” is skewed toward higher values, the “breathy” technique shows a much broader and flatter distribution across all features.



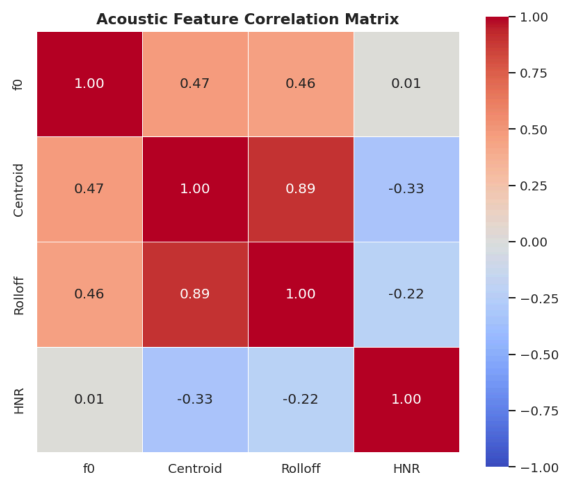
b. Boxplots to highlight that the “breathy” technique has significantly higher variance in f_0 and Spectral Rolloff than the “belt” technique.



c. Scatter Plot (f_0 vs. Centroid) to visualize the overlap between classes, suggesting that while “belt” tends to have a higher Centroid (brighter), there is no clear linear separation based on just these two features.



d. Heatmap used to quantify multicollinearity, highlighting the Centroid to Rolloff relationship.



5. Preliminary Insights and Hypothesis Formation

a. Based on the distinct two-cluster split in the f_0 distribution, we can assume the presence of a confounding variable, such as the difference in pitch for different genders. As a result, we'll have to adjust the model to learn the vocal technique regardless of fundamental pitch or normalize the features relative to the individual singers, if possible.

b. The 0.89 correlation between the Spectral Centroid and Rolloff features indicates that they provide highly redundant "brightness" information; therefore, assuming that multicollinearity can be introduced in linear models, we will likely drop one or adjust the dimensions during feature engineering.

c. The class overlap shown in the scatter plots and histograms indicates that a simple linear decision boundary would not be enough, so we'll aim to utilize non-linear machine learning classifiers, such as Random Forests, Support Vector Machines (SVM), or Gradient Boosting, better to capture the acoustic boundaries between the two vocal techniques.